20251204

Suppose we have a homogeneous scalar ODE of nth order

$$y^{(n)} + a_{n-1}(t)\, y^{(n-1)} + a_{n-2}(t)\, y^{(n-2)} + \cdots + a_1(t) y' + a_0(t) y = 0$$

we also have n sol'ns

$$y_1(t), \ldots, y_n(t)$$

we can construct an $n \times n$ matrix w/ determinant

$$W(t) = \det \begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}$$

called the Wronskian (determinant).

Abel's identity

$$\frac{d}{dt} W(t) = \overset{tr(A(t))}{\overbrace{-a_{n-1}(t)}}\, W(t)$$

The Wronskian is the same as that of the eqn

$$\frac{d}{dt}\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & 1 & & & O \\ & 0 & 1 & & \\ & & \ddots & \ddots & \\ & O & & 0 & 1 \\ -a_0(t) & -a_1(t) & \cdots & & -a_{n-1}(t) \end{bmatrix}}_{A(t)} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

where $x_1 = y,\ x_2 = y',\ \dots,\ x_n = y^{(n-1)}$.

Let's work out Abel's identity

$$\frac{d}{dt}\det\left(\begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}\right)$$

$$= \cancelto{0}{\det\begin{bmatrix} y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}}$$

$$+ \cancelto{0}{\det\begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}}$$

$$+ \cdots + \det\begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & \ddots & \vdots \\ y_1^{(n-2)}(t) & y_2^{(n-2)}(t) & \cdots & y^{(n-2)}(t) \\ y_1^{(n)}(t) & y_2^{(n)}(t) & \cdots & y_n^{(n)}(t) \end{bmatrix}$$

/

Note that $\forall k,\ 1 \le k \le n$

$$y_k^{(n)}(t) = -a_{n-1}(t)\, y_k^{(n-1)} - \cdots - a_1(t)\, y_k'(t) - a_0(t)\, y_k(t)$$

Then again by multilinearity

$$= \det \begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & & \vdots \\ -a_{n-1}(t)\, y_1^{(n-1)}(t) & -a_{n-1}(t)\, y_2^{(n-1)}(t) & \cdots & -a_{n-1}(t)\, y_n^{(n-1)}(t) \end{bmatrix}$$

$$= -a_{n-1}(t) \det \begin{bmatrix} y_1(t) & y_2(t) & \cdots & y_n(t) \\ y_1'(t) & y_2'(t) & \cdots & y_n'(t) \\ y_1''(t) & y_2''(t) & \cdots & y_n''(t) \\ \vdots & \vdots & & \vdots \\ y_1^{(n-1)}(t) & y_2^{(n-1)}(t) & \cdots & y_n^{(n-1)}(t) \end{bmatrix}$$

$$= -a_{n-1}(t)\, W(t).$$

As a result of Abel's identity

interval of sol'n

If $W(t_0) = 0$ for some $t_0 \in I$, then

$W(t) \equiv 0$ &

$y_1(t), \ldots, y_n(t)$ are linearly dependent

If $W(t_0) \neq 0$ for some $t_0 \in I$, then

$W(t) \neq 0 \ \forall t \in I$

$y_1(t), \ldots, y_n(t)$ are linearly independent

Consider the inhomogeneous ODE

$$\dot{x} = A(t)x + b(t) \quad , \quad x(t), b(t) \in \mathbb{R}^n \ \forall t$$

$A(t) \in M_n(\mathbb{R}) \ \forall t$

$A(\cdot)$ is continuous over some interval $I \subseteq \mathbb{R}$

Suppose

$$\Psi(t) = \begin{bmatrix} | & | & & | \\ x_1(t) & x_2(t) & \cdots & x_n(t) \\ | & | & & | \end{bmatrix}$$

is a fundamental matrix for the homogeneous eqn $\dot{x} = A(t)x$.

$x_1(t), \ldots, x_n(t) \in \mathbb{R}^n \ \forall t$ are linearly independent

Then

$$z(t) = \Psi(t) \int_{t_0}^{t} \Psi^{-1}(s)\, b(s)\, ds$$

is a particular sol'n.

Indeed,

$$\frac{d}{dt} z(t) = \frac{d}{dt}\left[\Psi(t) \int_{t_0}^{t} \Psi^{-1}(s)\, b(s)\, ds \right]$$

$$= \dot{\Psi}(t) \int_{t_0}^{t} \Psi^{-1}(s)\, b(s)\, ds + \cancel{\Psi(t)\, \Psi^{-1}(t)}\, b(t)$$

$$= A(t) \underbrace{\Psi(t) \int_{t_0}^{t} \Psi^{-1}(s)\, b(s)\, ds}_{z(t)} + b(t)$$

$$= A(t)\, z(t) + b(t) \qquad \square$$

Boyce & Diprima, thm 3.6.1, p 189

$$y'' + p(t) y' + q(t) y = g(t), \qquad t \in I \subseteq \mathbb{R}$$

$p, q, g \in C^0(I)$
continuous functions over I.

Suppose $y_1(t)$ & $y_2(t)$ are two independent sol'ns to the homogeneous eqn

$$y'' + p(t) y' + q(t) y = 0$$

and let $W(t) = \det \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix}$

Then a particular sol'n is given by

$$y_p(t) = -y_1(t) \int_{t_0}^{t} \frac{y_2(s)\, g(s)}{W(s)}\, ds + y_2(t) \int_{t_0}^{t} \frac{y_1(s)\, g(s)}{W(s)}\, ds$$

Essentially, we can convert the 2nd order ODE to

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 0 & 1 \\ -q(t) & -p(t) \end{pmatrix}}_{A(t)} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \underbrace{\begin{pmatrix} 0 \\ g(t) \end{pmatrix}}_{b(t)}$$

where $x_1 = y$, $x_2 = y'$.

so $$\Psi(t) = \begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix}$$

is a fund. matrix to the hom. eqn.

Therefore

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \Psi(t) \int_{t_0}^{t} \Psi(s)^{-1} b(s)\, ds$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{bmatrix} \int_{t_0}^{t} \begin{bmatrix} -\dfrac{y_2(s)\, g(s)}{W(s)} \\ \dfrac{y_1(s)\, g(s)}{W(s)} \end{bmatrix} ds$$

$$= \begin{bmatrix} y_1(t) & y_2(t) \end{bmatrix} \begin{bmatrix} \displaystyle\int_{t_0}^{t} -\frac{y_2(s)\, g(s)}{W(s)}\, ds \\ \displaystyle\int_{t_0}^{t} \frac{y_1(s)\, g(s)}{W(s)}\, ds \end{bmatrix}$$

$$= -y_1(t) \int_{t_0}^{t} \frac{y_2(s) g(s)}{W(s)}\, ds + y_2(t) \int_{t_0}^{t} \frac{y_1(s) g(s)}{W(s)}\, ds \quad \checkmark$$

where

$$\Psi(s)^{-1} b(s) = \begin{bmatrix} y_1(s) & y_2(s) \\ y_1'(s) & y_2'(s) \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ g(s) \end{bmatrix}$$

$$= \begin{bmatrix} \det \begin{pmatrix} 0 & y_2(s) \\ g(s) & y_2'(s) \end{pmatrix} \Big/ \det \begin{pmatrix} y_1(s) & y_2(s) \\ y_1'(s) & y_2'(s) \end{pmatrix} \\ \det \begin{pmatrix} y_1(s) & 0 \\ y_1'(s) & g(s) \end{pmatrix} \Big/ \det \begin{pmatrix} y_1(s) & y_2(s) \\ y_1'(s) & y_2'(s) \end{pmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} -\dfrac{y_2(s)\, g(s)}{W(s)} \\ \dfrac{y_1(s)\, g(s)}{W(s)} \end{bmatrix}$$

For higher order results, similarly c.f.

Wolfgang Walter, p. 202-203

Gradient of scalar-valued matrix function.

We would like to compute the gradient of a certain scalar-valued function $f : V \longrightarrow \mathbb{R}$

V: vector space

e.g., $\det : M_n(\mathbb{R}) \longrightarrow \mathbb{R}$

Consider $f : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}$

$$(x_1, x_2, x_3) \longmapsto f(x_1, x_2, x_3)$$

or $\quad x \longmapsto f(x)$ where $x = (x_1, x_2, x_3) \in \mathbb{R}^3$

$$= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3$$

Suppose $f \in C^1$, then the gradient of f at $x \in D$ is a column vector $\nabla f(x) \in \mathbb{R}^3$, so

$$\nabla f : D \subseteq \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$$x \longmapsto \nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{bmatrix} \Bigg|_{x=(x_1,x_2,x_3)}$$

$$\left(= \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{bmatrix} (x) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x) \\ \frac{\partial f}{\partial x_2}(x) \\ \frac{\partial f}{\partial x_3}(x) \end{bmatrix} \right)$$

On the hand, the Jacobian of f at x is given by a row vector

1×3

$$Df(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \frac{\partial f}{\partial x_3} \end{bmatrix} \Bigg|_{x=(x_1,x_2,x_3)} : \mathbb{R}^3 \to \mathbb{R}$$

$$h \longmapsto Df(x) \cdot h = \frac{\partial f}{\partial x_1} h_1 + \frac{\partial f}{\partial x_2} h_2 + \frac{\partial f}{\partial x_3} h_3$$

$$=: \langle \nabla f(x), h \rangle$$ inner product in $\mathbb{R}^3$

$$= \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \frac{\partial f}{\partial x_3} \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

$$\phi$$

$$\langle x, y \rangle := \sum_{i=1}^{3} x_i y_i$$

we can define

$$\nabla f(x) := \left(D f(x) \right)^T$$

Consider inner product over $M_n(\mathbb{R})$

$$\langle \cdot , \cdot \rangle : M_n(\mathbb{R}) \times M_n(\mathbb{R}) \longrightarrow \mathbb{R}$$

$$(A, B) \longmapsto tr(A^T B) = tr(BA^T) = tr(B^T A)$$

$$= tr(AB^T)$$

$$= \sum_{i,j=1}^{n} a_{ij} b_{ij}$$

Then given $f : U \subseteq M_n(\mathbb{R}) \to \mathbb{R}$

The gradient of f at $A \in U$, denoted by $\nabla f(A)$, is given by

$$Df(A) \cdot H = \langle \nabla f(A), H \rangle \quad \forall H \in M_n(\mathbb{R})$$

$$= tr\left((\nabla f(A))^T H \right)$$

$$= tr\left(\nabla f(A) H^T \right)$$

Ex $f : GL_n(\mathbb{R}) \to \mathbb{R}$, $f(A) = \log(\det(A^{-1}))$

$$Df(A)\cdot H = D\log(\det(A^{-1}))\cdot D\det(A^{-1})\cdot \underbrace{Di(A)\cdot H}$$

$$= D\log(\det(A^{-1}))\cdot \underbrace{D\det(A^{-1})\cdot -A^{-1}HA^{-1}}$$

$$= \underbrace{D\log(\det(A^{-1}))\cdot \det(A^{-1})\, tr(\cancel{(A^{-1})^{-1}}(-\cancel{A^{-1}}HA^{-1}))}$$

$$= \frac{1}{\cancel{\det(A^{-1})}}\cancel{\det(A^{-1})}\, tr(-HA^{-1})$$

$$= tr(-A^{-1}H) = tr((-A^{-T})^{T}H) = \langle -A^{-T}, H\rangle$$

$$\Rightarrow \nabla f(A) = -A^{-T}$$

Ex $f(A) = tr(e^{A})$, $A \in M_n(\mathbb{R})$

$$Df(A)\cdot H = D(tr\circ exp)(A)\cdot H$$

$$= Dtr(e^{A})\cdot Dexp(A)\cdot H$$

$$= tr(Dexp(A)\cdot H) \quad \text{(linearity of trace)}$$

$$= tr(e^{A}H) \quad \text{(HW)}$$

$$= tr((e^{A^T})^{T}H)$$

$$= \langle e^{A^T}, H\rangle$$

$$\Rightarrow \nabla f(A) = e^{A^T} = (e^{A})^{T}.$$

$$Dtr(A)\cdot H$$

$$= \frac{d}{dt}\Big|_{t=0} tr(A+tH)$$

$$= \frac{d}{dt}\Big|_{t=0} (tr(A)+t\,tr(H))$$

$$= tr(H)$$

indep. of A.